

Canonical ensemble methods in Thermal-FIST

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1 Overview

The `ThermalModelCanonical` class computes thermodynamics of a hadron resonance gas (HRG) in the canonical ensemble (CE) with exact conservation of baryon number B , electric charge Q , strangeness S , and optionally charm C . The CE partition function reads

$$Z(\mathbf{N}) = \int_{-\pi}^{\pi} \frac{d\phi_B}{2\pi} \int_{-\pi}^{\pi} \frac{d\phi_Q}{2\pi} \int_{-\pi}^{\pi} \frac{d\phi_S}{2\pi} e^{g(\phi)}, \quad (1)$$

where $\mathbf{N} = (B, Q, S, \dots)$ are the conserved charges and

$$g(\phi) = -i \mathbf{N} \cdot \phi + \sum_j \sum_{n=1}^{n_{\max}} z_j^n e^{i n \mathbf{q}_j \cdot \phi}. \quad (2)$$

Mean multiplicities and covariances are obtained from partition function ratios $R(\mathbf{c}) \equiv Z(\mathbf{N} - \mathbf{c})/Z(\mathbf{N})$:

$$\langle N_j \rangle_{\text{CE}} = \sum_n n z_j^n R(n \mathbf{q}_j), \quad (3)$$

$$\text{cov}_{\text{CE}}(N_j, N_k) = \delta_{jk} \sum_n n^2 z_j^n R(n \mathbf{q}_j) + \sum_{n,m} n m z_j^n z_k^m \left[R(n \mathbf{q}_j + m \mathbf{q}_k) - R(n \mathbf{q}_j) R(m \mathbf{q}_k) \right]. \quad (4)$$

Four methods are available, selected via `SetMethod()`:

<code>CanonicalMethod</code>	Value	Description
<code>GaussLegendre</code>	0	Multi-dimensional numerical quadrature
<code>SaddlePoint</code>	1	Per-quantum-number saddle-point solve
<code>SaddlePointNLO</code>	2	Analytic LO+NLO from CGF expansion
<code>SaddlePointLO</code>	3	Analytic LO only from CGF expansion

2 Gauss–Legendre quadrature (`GaussLegendre`)

The multi-dimensional Fourier integral (1) is evaluated by composite 10-point Gauss–Legendre quadrature on $[0, \pi]^d$ (d = number of canonical charges). The interval for each angle is divided into $n_{\max}$ subintervals, with

$$n_{\max} = m \times \max\left(3, \sqrt{B^2 + Q^2 + S^2 + C^2}\right), \quad (5)$$

where $m \geq 1$ is a user-adjustable multiplier (`SetIntegrationIterationsMultiplier()`).

Baryon analytical integration. When enabled (`m_Banalyt`), the baryon fugacity ϕ_B is integrated analytically using modified Bessel functions, reducing the quadrature dimension by one:

$$\int_0^\pi \frac{d\phi_B}{\pi} \cos(B_g \phi_B) \exp\left(2|w| \cos(\phi_B - \psi)\right) = I_{B_g}(2|w|). \quad (6)$$

Saddle-point contour shift. When the baryon analytical path is *not* active (`m_Banalyt = false`), the code shifts the Fourier integration contour through the saddle point before evaluating the GL quadrature. Concretely, `PrepareModelGCE()` solves the GCE constraint equations

$$\sum_j \sum_n n q_{a,j} \omega_j^{*,n} = N_a \quad (a \in \text{canonical charges}) \quad (7)$$

for the saddle-point chemical potentials μ_{sp} , and `FillChemicalPotentials()` propagates them to each species. This is equivalent to the contour deformation $\phi_a \rightarrow \phi_a + i\mu_{\text{sp},a}/T$ (Cauchy's theorem for periodic analytic functions) and is mathematically exact.

The effect is to populate the antisymmetric multiplicities $\text{Nsy}[\mathbf{q}] = V_c \sum_{j:\mathbf{q}_j=\mathbf{q}} \sinh(\mu_j/T) z_j$ in the integrand (Eq. 2). At the saddle point, the condition $\sum_{\mathbf{q}} q_a \text{Nsy}[\mathbf{q}] = N_a$ ensures that the linear phase term $-N_a \phi_a + \sum_{\mathbf{q}} q_a \text{Nsy}[\mathbf{q}] \sin(q_a \phi_a)$ vanishes at $\phi = 0$ to first order. This eliminates the rapid oscillation that occurs at finite chemical potentials (e.g. strangeness-canonical at nonzero μ_B), where the large Nsy terms from strange baryons cause severe cancellation in the integrand.

The multiplicative prefactor $e^{-\mathbf{N} \cdot \mu_{\text{sp}}/T}$ introduced by the contour shift cancels implicitly: in the partition function ratios $R(\mathbf{c}) = Z(\mathbf{N} - \mathbf{c})/Z(\mathbf{N})$, it leaves a residual factor $e^{\mathbf{c} \cdot \mu_{\text{sp}}/T}$ that is exactly compensated by the shifted GCE densities $n_j^{\text{GCE}}(\mu_{\text{sp}})$ already used in $\langle N_j \rangle_{\text{CE}} = R(\mathbf{q}_j) n_j^{\text{GCE}}$. For entropy, the shift contributes $\ln Z(\mathbf{N}) = -\mathbf{N} \cdot \mu_{\text{sp}}/T + \ln \tilde{Z}(\mathbf{N})$; the additional term $-\mu_{\text{sp}} \cdot \mathbf{N}/(T V_c)$ combines with the shifted thermal contributions to reproduce the correct canonical entropy. No explicit prefactor correction is needed for any thermodynamic observable.

The contour shift is *not* applied in the `m_Banalyt` mode. The Bessel identity used there assumes particle–antiparticle symmetry ($\text{Nsx}[\mathbf{q}] = \text{Nsx}[-\mathbf{q}]$), which breaks at $\mu \neq 0$. This restriction is benign: `m_Banalyt` is active only when all four charges are canonical, where the integrand at $\mu = 0$ is already symmetric under $\phi \rightarrow -\phi$ and has no oscillation problem.

Outputs. The quadrature produces $Z(\mathbf{N} - \mathbf{c})$ for all needed charge vectors $\mathbf{c}$. The ratios $R(\mathbf{c})$ are then fed into Eqs. (3)–(4) to obtain means and covariances.

Limitations. At large correlation volumes ($V_c \gtrsim 5000 \text{ fm}^3$), catastrophic cancellation in double precision degrades the fluctuation observables: $R(\mathbf{c} + \mathbf{c}')/[R(\mathbf{c}) R(\mathbf{c}')]$ deviates from unity by only $\mathcal{O}(V_c^{-1})$ but both numerator and denominator are $\sim e^{\mathcal{O}(V_c)}$. The saddle-point contour shift eliminates the oscillation problem at finite μ but does not resolve the peak-narrowness issue at very large volumes: the integrand peak width scales as $\sim 1/\sqrt{N_{\text{tot}}}$, which can be too narrow for the GL quadrature to resolve even with the contour shift. In this regime, the saddle-point methods (Sections 3–4) should be used instead.

3 Saddle-point approximation (SaddlePoint)

This method applies the Gaussian saddle-point approximation to each partition function individually, then forms ratios.

3.1 Saddle-point ingredients

Deform the contour $\phi_a = \mu_a/T + i\theta_a$ and define:

$$\omega_j^{*,n} \equiv z_j^n \exp\left(n \mathbf{q}_j \cdot \frac{\boldsymbol{\mu}^*}{T}\right), \quad (8)$$

where $\boldsymbol{\mu}^*/T$ solves the saddle-point (GCE constraint) equations:

$$\sum_j \sum_n n q_{a,j} \omega_j^{*,n} = N_a, \quad a \in \{B, Q, S, \dots\}. \quad (9)$$

The *susceptibility matrix* is

$$\Sigma_{ab} = \sum_j q_{a,j} q_{b,j} W_j^{(2)}, \quad W_j^{(k)} \equiv \sum_n n^k \omega_j^{*,n}. \quad (10)$$

3.2 Partition function ratios

For each charge vector $\mathbf{c}$, solve the shifted saddle-point equations for $\boldsymbol{\mu}_{\mathbf{c}}^*$ with charges $\mathbf{N} - \mathbf{c}$ and compute the ratio as

$$\ln R(\mathbf{c}) = \frac{V_c}{T} [p(\boldsymbol{\mu}_{\mathbf{c}}^*) - p(\boldsymbol{\mu}_0^*)] - \left[\sum_a \frac{\mu_{\mathbf{c},a}^* (N_a - c_a)}{T} - \sum_a \frac{\mu_{0,a}^* N_a}{T} \right] - \frac{1}{2} [\ln \det \Sigma_{\mathbf{c}} - \ln \det \Sigma_0], \quad (11)$$

where $p(\boldsymbol{\mu}) = \sum_j p_j(T, \mu_j^*)$ is the total GCE pressure and $\Sigma_{\mathbf{c}}$ is evaluated at $\boldsymbol{\mu}_{\mathbf{c}}^*$. Each of the three terms is $\mathcal{O}(1)$, avoiding the catastrophic cancellation that plagues GL at large V_c .

3.3 Algorithm

1. Solve Eq. (9) for $\boldsymbol{\mu}_0^*$ and compute p_0 , Σ_0 , $\ln \det \Sigma_0$.
2. For each $\mathbf{c} \neq 0$: solve the shifted equations via Broyden's method (warm-started from the previous solution), compute $p_{\mathbf{c}}$, $\Sigma_{\mathbf{c}}$, and evaluate Eq. (11).
3. Feed the $R(\mathbf{c})$ values into Eqs. (3)–(4).

Conservation laws. Mean charges $\sum_j q_{a,j} \langle N_j \rangle_{\text{CE}}$ deviate from N_a by $\mathcal{O}(V_c^{-1})$, since the Gaussian saddle-point for each $R(\mathbf{c})$ is evaluated independently.

4 Saddle-point NLO (SaddlePointNLO)

This method computes means and covariances directly from the saddle-point expansion of the cumulant generating function (CGF), bypassing partition function ratios entirely. See the companion note [1] for full derivations.

4.1 Key quantities

At the saddle point $\boldsymbol{\mu}^*/T$ (Eq. 9), define:

- Cluster moments: $W_j^{(k)} = \sum_n n^k \omega_j^{*,n}$ ($k = 1, 2, 3, 4$).
- Susceptibility matrix: $\Sigma_{ab} = \sum_j q_{a,j} q_{b,j} W_j^{(2)}$.
- Projection: $G_{jk} = \mathbf{q}_j^T \Sigma^{-1} \mathbf{q}_k$ [$\mathcal{O}(V_c^{-1})$].
- Response matrix: $D_{jm} = \delta_{jm} - W_m^{(2)} G_{jm}$.

4.2 Mean yields (LO + NLO)

$$\boxed{\langle N_l \rangle_{\text{CE}} = W_l^{(1)} - \frac{1}{2} W_l^{(3)} G_{ll} + \frac{1}{2} W_l^{(2)} \sum_j W_j^{(3)} G_{jj} G_{jl} + \mathcal{O}(V_c^{-1})}. \quad (12)$$

The first term is the GCE mean [$\mathcal{O}(V_c)$]. The second and third are the NLO canonical correction [$\mathcal{O}(V_c^0)$], involving the third cluster moment $W^{(3)}$ (quantum-statistical skewness).

4.3 Covariances (LO + NLO)

$$\text{cov}_{\text{CE}}(N_l, N_m) = \underbrace{\delta_{lm} W_l^{(2)} - W_l^{(2)} W_m^{(2)} G_{lm}}_{\text{LO: } \mathcal{O}(V_c)} - \underbrace{\frac{1}{2} B_{lm} + \frac{1}{2} C_{lm}}_{\text{NLO: } \mathcal{O}(1)} + \mathcal{O}(V_c^{-1}), \quad (13)$$

where the NLO terms involve auxiliary quantities:

$$C_{lm} = \text{Tr}(M_l M_m), \quad (M_l)_{ca} = \sum_j W_j^{(3)} D_{jl} P_{j,c} q_{a,j}, \quad P_{j,a} = \sum_b (\Sigma^{-1})_{ab} q_{b,j}, \quad (14)$$

$$B_{lm} = \sum_j G_{jj} \left[W_j^{(4)} D_{jl} D_{jm} + W_j^{(3)} \frac{\partial^2 \tilde{\alpha}_j}{\partial \alpha_l \partial \alpha_m} \Big|_0 \right], \quad (15)$$

with the second-derivative response

$$\frac{\partial^2 \tilde{\alpha}_j}{\partial \alpha_l \partial \alpha_m} \Big|_0 = - \sum_a q_{a,j} \sum_b (\Sigma^{-1})_{ab} S_{b,lm}, \quad S_{b,lm} = \sum_k q_{b,k} W_k^{(3)} D_{kl} D_{km}. \quad (16)$$

4.4 Thermodynamic quantities (LO + NLO)

Bulk thermodynamic quantities—pressure, energy density, and entropy density—follow from derivatives of $\ln Z(\mathbf{N})$ with respect to the thermodynamic variables. We derive each in turn, starting from the saddle-point expression for the canonical partition function.

4.4.1 Starting point: saddle-point $\ln Z$

After performing the Gaussian integral around the saddle point $\boldsymbol{\mu}^*/T$, the canonical partition function is

$$\ln Z(\mathbf{N}) = \frac{V_c}{T} p^{\text{GCE}}(\boldsymbol{\mu}^*) - \sum_a \frac{\mu_a^* N_a}{T} - \frac{d}{2} \ln(2\pi) - \frac{1}{2} \ln \det \Sigma + \mathcal{O}(V_c^{-1}), \quad (17)$$

where $p^{\text{GCE}}(\boldsymbol{\mu}^*) = T \sum_j \sum_n \omega_j^{*,n} / V_c$ is the GCE pressure at the saddle-point chemical potentials, d is the number of canonically conserved charges with nonzero susceptibility (i.e. $d = \text{rank}(\Sigma)$), and Σ_{ab} is defined in Eq. (10).

Stationarity. The saddle-point condition (9) ensures $\partial(\ln Z)/\partial\alpha_a^* = 0$ (where $\alpha_a \equiv \mu_a/T$). This means that when differentiating $\ln Z$ with respect to external parameters (T , V_c), the implicit dependence $\boldsymbol{\mu}^*(T, V_c)$ drops out, and only the *explicit* dependence on T and V_c contributes. This is the key simplification.

4.4.2 Pressure

The canonical pressure is

$$P = T \left. \frac{\partial \ln Z(\mathbf{N})}{\partial V_c} \right|_T. \quad (18)$$

From Eq. (17), the V_c -dependent terms are:

- $V_c p^{\text{GCE}}/T$: Since $\omega_j^{*,n} \propto V_c$ at fixed $\alpha_a^* = \mu_a^*/T$, the GCE pressure p^{GCE} is V_c -independent, so $\partial(V_c p^{\text{GCE}}/T)/\partial V_c = p^{\text{GCE}}/T$.
- $-\sum_a \mu_a^* N_a/T$: By stationarity, this is constant (the saddle-point $\boldsymbol{\mu}^*$ does not change with V_c at fixed $\alpha^* = \mu^*/T$, and N_a is fixed).
- $-\frac{1}{2} \ln \det \Sigma$: Since $\Sigma_{ab} = \sum_j q_{a,j} q_{b,j} W_j^{(2)}$ and each $W_j^{(2)} = \sum_n n^2 \omega_j^{*,n} \propto V_c$ at fixed α^* , we have $\Sigma = V_c \hat{\Sigma}$ where $\hat{\Sigma}$ is V_c -independent. Thus $\ln \det \Sigma = d \ln V_c + \ln \det \hat{\Sigma}$, and $\partial \ln \det \Sigma / \partial V_c = d/V_c$.

Combining:

$$P = p^{\text{GCE}}(\boldsymbol{\mu}^*) - \frac{dT}{2V_c} + \mathcal{O}(V_c^{-2}). \quad (19)$$

The NLO correction $-dT/(2V_c)$ is purely geometric (it counts the number of conserved-charge constraints) and is independent of the particle content.

4.4.3 Energy density

The canonical energy density is

$$\varepsilon = \frac{T^2}{V_c} \left. \frac{\partial \ln Z(\mathbf{N})}{\partial T} \right|_{V_c}. \quad (20)$$

We differentiate Eq. (17) with respect to T at fixed V_c , using stationarity to ignore the implicit T -dependence through $\boldsymbol{\mu}^*(T)$. Writing $\alpha_a^* = \mu_a^*/T$, the “fixed α^* ” derivatives are:

- $\partial(V_c p^{\text{GCE}}/T)/\partial T|_{\alpha^*} = V_c \varepsilon^{\text{GCE}}/T^2$, from the identity $T^2 \partial(p/T)/\partial T|_{\alpha} = \varepsilon$ (derived below, Eq. 25).
- $\partial(-\sum_a \alpha_a^* N_a)/\partial T|_{\alpha^*} = 0$ (fixed).
- $-\frac{1}{2} \partial \ln \det \Sigma / \partial T|_{\alpha^*} = -\frac{1}{2} \text{Tr}(\Sigma^{-1} \partial \Sigma / \partial T|_{\alpha^*})$.

The first two terms give the LO contribution $\varepsilon^{\text{GCE}}(\mu^*)$. The NLO correction requires computing $\partial \Sigma / \partial T|_{\alpha^*}$.

Temperature derivative of Σ . Since $\Sigma_{ab} = \sum_j q_{a,j} q_{b,j} W_j^{(2)}$ and $W_j^{(2)} = \sum_n n^2 \omega_j^{*,n}$, we need $\partial \omega_j^{*,n} / \partial T|_{\alpha^*}$. From the definition (8), $\omega_j^{*,n} = V_c p_{j,n} / T$, where $p_{j,n}$ is the n -th cluster contribution to the GCE pressure of species j . The standard cluster expansion gives

$$p_{j,n} = \frac{(\pm 1)^{n+1}}{n} \frac{d_j T}{2\pi^2} m_j^2 K_2\left(\frac{n m_j}{T}\right) e^{n\mu_j/T} = (\pm 1)^{n+1} p_j^{\text{Boltz}}\left(\frac{T}{n}, \mu_j\right), \quad (21)$$

where the second equality holds because the Boltzmann pressure at temperature T/n has a prefactor $d_j (T/n)/(2\pi^2) = d_j T/(n \cdot 2\pi^2)$, which precisely absorbs the cluster-expansion factor $1/n$. This is what `DensityCluster( $n$ , Pressure)` returns.

We define $\varepsilon_{j,n}$ through the same relation used for the total pressure:

$$\varepsilon_{j,n} \equiv T^2 \frac{\partial(p_{j,n}/T)}{\partial T} \Big|_{\alpha}. \quad (22)$$

Using the identity (proved below) $T^2 \partial(p/T)/\partial T|_{\alpha} = \varepsilon$, this gives

$$\frac{\partial \omega_j^{*,n}}{\partial T} \Big|_{\alpha^*} = \frac{V_c}{T^2} \varepsilon_{j,n}(\mu_j^*). \quad (23)$$

One can verify that $\varepsilon_{j,n} = (\pm 1)^{n+1} \varepsilon_j^{\text{Boltz}}(T/n, \mu_j)$, which is what `DensityCluster( $n$ , EnergyDensity)` returns (see below).

Proof of the identity $T^2 \partial(p/T)/\partial T|_{\alpha} = \varepsilon$. One might naively expect the entropy density here, since $\partial p / \partial T|_{\mu} = s$. The key is that we differentiate p/T (not p) and hold $\alpha = \mu/T$ fixed (not μ). Using $\mu = \alpha T$ and the chain rule:

$$\frac{\partial p}{\partial T} \Big|_{\alpha} = \frac{\partial p}{\partial T} \Big|_{\mu} + \frac{\partial p}{\partial \mu} \Big|_T \frac{\mu}{T} = s + \frac{n\mu}{T}, \quad (24)$$

where s and n are the entropy and number densities. Then:

$$\frac{\partial(p/T)}{\partial T} \Big|_{\alpha} = \frac{1}{T} \frac{\partial p}{\partial T} \Big|_{\alpha} - \frac{p}{T^2} = \frac{Ts + n\mu - p}{T^2} = \frac{\varepsilon}{T^2}, \quad (25)$$

where the last step uses the Euler relation $\varepsilon + p = Ts + \mu n$.

Proof that $\varepsilon_{j,n} = (\pm 1)^{n+1} \varepsilon_j^{\text{Boltz}}(T/n, \mu_j)$. From Eq. (21), $p_{j,n}/T = (\pm 1)^{n+1} p_j^{\text{Boltz}}(T/n, \mu_j)/T$. Let $\tilde{T} = T/n$, so $T = n\tilde{T}$. At fixed $\alpha = \mu/T$, the ratio $\mu/\tilde{T} = n\alpha$ is also constant. Hence $\partial/\partial T|_{\alpha} = (1/n) \partial/\partial \tilde{T}|_{\mu/\tilde{T}}$ and

$$T^2 \frac{\partial(p^{\text{Boltz}}(\tilde{T}, \mu)/(n\tilde{T}))}{\partial T} \Big|_{\alpha} = \frac{n^2 \tilde{T}^2}{n} \cdot \frac{1}{n} \frac{\partial(p^{\text{Boltz}}/\tilde{T})}{\partial \tilde{T}} \Big|_{\mu/\tilde{T}} = \tilde{T}^2 \frac{\partial(p^{\text{Boltz}}/\tilde{T})}{\partial \tilde{T}} \Big|_{\mu/\tilde{T}} = \varepsilon^{\text{Boltz}}(\tilde{T}, \mu), \quad (26)$$

where the last equality uses Eq. (25) for a Boltzmann gas at temperature $\tilde{T}$. Therefore $\varepsilon_{j,n} = (\pm 1)^{n+1} \varepsilon_j^{\text{Boltz}}(T/n, \mu_j)$, confirming that `DensityCluster(n, EnergyDensity)` directly gives $\varepsilon_{j,n}$.
Therefore:

$$\frac{\partial W_j^{(2)}}{\partial T} \Big|_{\alpha^*} = \sum_n n^2 \frac{\partial \omega_j^{*,n}}{\partial T} \Big|_{\alpha^*} = \frac{V_c}{T^2} \sum_n n^2 \varepsilon_{j,n}(\mu_j^*). \quad (27)$$

Define the *energy-weighted cluster sum*:

$$E_j \equiv \sum_{n=1}^{n_{\max}} n^2 \varepsilon_{j,n}(\mu_j^*). \quad (28)$$

The n^2 factor arises because $W_j^{(2)} = \sum_n n^2 \omega_j^{*,n}$, so differentiating brings down n^2 times $\partial\omega/\partial T$. For Boltzmann statistics only $n = 1$ contributes, giving $E_j = \varepsilon_j^{\text{GCE}}(\mu_j^*)$.

Trace formula. Substituting into the susceptibility derivative:

$$\frac{\partial \Sigma_{ab}}{\partial T} \Big|_{\alpha^*} = \sum_j q_{a,j} q_{b,j} \frac{V_c E_j}{T^2}, \quad (29)$$

and using $G_{jk} = \mathbf{q}_j^T \Sigma^{-1} \mathbf{q}_k$:

$$\text{Tr} \left(\Sigma^{-1} \frac{\partial \Sigma}{\partial T} \Big|_{\alpha^*} \right) = \sum_j G_{jj} \frac{V_c E_j}{T^2}. \quad (30)$$

The NLO energy correction is thus

$$\delta\varepsilon = -\frac{T^2}{2V_c} \text{Tr} \left(\Sigma^{-1} \frac{\partial \Sigma}{\partial T} \Big|_{\alpha^*} \right) = -\frac{1}{2} \sum_j E_j G_{jj}, \quad (31)$$

giving the full result:

$$\varepsilon = \varepsilon^{\text{GCE}}(\mu^*) - \frac{1}{2} \sum_j E_j G_{jj} + \mathcal{O}(V_c^{-2}), \quad (32)$$

where $\varepsilon^{\text{GCE}}(\mu^*) = \sum_j \varepsilon_j(T, \mu_j^*)$ is the GCE energy density at the saddle point.

4.4.4 Entropy density

The canonical entropy density follows from the exact statistical mechanics identity (no chemical potential term in the canonical ensemble since the charges are fixed):

$$s = \frac{\ln Z(\mathbf{N})}{V_c} + \frac{\varepsilon}{T}. \quad (33)$$

This holds at any volume. Here $\ln Z(\mathbf{N})$ is given by the saddle-point expression (17), already computed and stored in `m_MultExp`. The energy density ε is given by Eq. (32). No separate NLO formula is needed—the entropy is automatically consistent through $\ln Z$ and ε .

Euler relation. The Euler relation $P = Ts - \varepsilon$ (equivalently $PV_c = T \ln Z$) holds only in the thermodynamic limit $V_c \rightarrow \infty$. At finite V_c , the deviation $s - (\varepsilon + P)/T = \ln Z/V_c - P/T$ is $\mathcal{O}(V_c^{-1})$. This can be verified explicitly: using Eq. (17),

$$\frac{\ln Z}{V_c} = \frac{p^{\text{GCE}}}{T} - \frac{\boldsymbol{\mu}^* \cdot \mathbf{N}}{TV_c} - \frac{d \ln(2\pi)}{2V_c} - \frac{\ln \det \Sigma}{2V_c}, \quad (34)$$

while $P/T = p^{\text{GCE}}/T - d/(2V_c)$, so the difference is $\mathcal{O}(V_c^{-1})$ as expected.

4.4.5 Numerical validation

The NLO corrections to pressure and energy density have been validated numerically by comparing against exact Gauss–Legendre (GL) quadrature results at moderate volumes.

Method. For a range of correlation volumes V_c , the GL method (Section 2) provides numerically exact results (up to quadrature error). At large enough V_c , all thermodynamic densities admit a $1/V_c$ expansion:

$$f(V_c) = f_{\text{LO}} + \frac{f_{\text{NLO}}}{V_c} + \mathcal{O}(V_c^{-2}). \quad (35)$$

The NLO coefficient f_{NLO} can be extracted from GL data via linear regression of f against $1/V_c$. Independently, the analytic NLO prediction is obtained from two `SaddlePointNLO` calculations at well-separated volumes V_1 and V_2 :

$$f_{\text{NLO}}^{(\text{analytic})} = \frac{V_1 V_2 [f(V_1) - f(V_2)]}{V_2 - V_1}. \quad (36)$$

Results: $d = 3$ (**no charmed hadrons**). Using the PDG2020 particle list without charmed hadrons at $T = 155$ MeV, $\mathbf{N} = 0$, with canonical B, Q, S, C conservation and Boltzmann statistics, GL calculations at $V_c = 500\text{--}2000 \text{ fm}^3$ with integration multiplier $m = 3$ give:

Quantity	GL slope	NLO slope	Ratio	Analytic
ε	agrees	agrees	≈ 0.99	—
P	≈ -0.233	≈ -0.233	≈ 0.99	$-3T/2 = -0.2325$

The pressure slope matches the analytic prediction $-dT/2$ with $d = 3$, corresponding to $\text{rank}(\Sigma) = 3$: although charm is formally canonical, the particle list contains no charmed hadrons, so $\Sigma_{CC} = 0$ and the charm direction does not contribute.

Results: $d = 4$ (**with charmed hadrons**). Switching to the particle list with charmed hadrons and setting a large charm fugacity $\gamma_C \gg 1$ to ensure $\Sigma_{CC} V_c \gg 1$, the GL pressure slope at $V_c = 200\text{--}1000 \text{ fm}^3$ gives ≈ -0.297 , consistent with the analytic prediction $-4T/2 = -0.310$. At physical $\gamma_C = 1$, where charm production is thermally suppressed ($\Sigma_{CC} V_c \ll 1$), the GL slope reverts to ≈ -0.240 , closer to $d = 3$.

Effective dimensionality. The parameter d in the NLO pressure formula (19) is the number of charge directions for which the saddle-point (Gaussian) approximation is valid. The Gaussian approximation for charge a requires that the integration range $[-\pi, \pi]$ be much wider than the Gaussian width $\sigma_a \sim 1/\sqrt{\Sigma_{aa} V_c}$, i.e.

$$\Sigma_{aa} V_c \gg 1. \quad (37)$$

When this condition is violated (as for charm at physical γ_C), the Gaussian extends well beyond $[-\pi, \pi]$ and the saddle-point approximation breaks down for that direction. The NLO formula then overestimates the correction from that charge: it predicts a $-T/(2V_c)$ contribution per charge, but the true correction is exponentially small when $\Sigma_{aa} V_c \rightarrow 0$. In practice, the effective dimensionality is $d = \text{rank}_{\text{eff}}(\Sigma V_c)$, counting only the directions satisfying (37).

Entropy limitation. The NLO entropy density (33) inherits a limitation from the saddle-point $\ln Z$ expression (17). The NLO terms in ε and P are correctly captured because they arise from derivatives of $\ln Z$ with respect to T and V_c , which naturally select the $\mathcal{O}(V_c^0)$ and $\mathcal{O}(V_c^{-1})$ contributions. However, $\ln Z$ itself contains an $\mathcal{O}(1)$ NNLO correction (from the next term in the saddle-point expansion) that is not computed. Since $s = \ln Z/V_c + \varepsilon/T$, this missing $\mathcal{O}(1)$ term in $\ln Z$ produces an $\mathcal{O}(V_c^{-1})$ error in s . This is a limitation of both the `SaddlePoint` and `SaddlePointNLO` methods—they share the same saddle-point $\ln Z$ —and does not affect ε or P .

4.5 Algorithm

1. Solve Eq. (9) for μ^*/T via Broyden’s method.
2. Compute $W_j^{(k)}$ ($k = 1, \dots, 4$), the GCE energy density $\varepsilon^{\text{GCE}}(\mu^*)$ and pressure $p^{\text{GCE}}(\mu^*)$, and the energy-weighted cluster sums E_j for all species from the cluster expansion.
3. Build $\Sigma_{ab} = \sum_j q_{a,j} q_{b,j} W_j^{(2)}$ directly from the $W_j^{(2)}$ values. Invert to get Σ^{-1} .
4. Compute G_{jj} , $P_{j,a}$, and D_{jl} for all species.
5. Evaluate NLO thermodynamic quantities from Eqs. (19)–(32).
6. Evaluate means from Eq. (12): cost $\mathcal{O}(N_{\text{spec}}^2)$.
7. If fluctuations are requested, evaluate the full covariance matrix from Eqs. (13)–(16): cost $\mathcal{O}(N_{\text{spec}}^3)$.

Conservation laws. The identity $\sum_l q_{a,l} D_{jl} = 0$ ensures $\sum_l q_{a,l} \langle N_l \rangle_{\text{CE}} = N_a$ exactly, order by order. Similarly, $\text{var}_{\text{CE}}(Q_a) = 0$ is satisfied at LO by $\Sigma - \Sigma \Sigma^{-1} \Sigma = 0$.

5 Selective canonical treatment

Any subset of charges can be treated canonically while the rest remain grand-canonical. For example, canonical B and S with grand-canonical Q : the saddle-point integral becomes two-dimensional, with a 2×2 susceptibility submatrix. All formulas hold with Σ^{-1} restricted to the canonical subspace and charge vectors projected accordingly.

6 Comparison

	GaussLegendre	SaddlePoint	SaddlePointNLO	SaddlePointLO
Partition functions	GL quadrature	per- $\mathbf{c}$ SP solve	not computed	not computed
Mean yields	from $R(\mathbf{c})$	from $R(\mathbf{c})$	direct (Eq. 12)	GCE at μ^*
Fluctuations	from $R(\mathbf{c} + \mathbf{c}')$	from $R(\mathbf{c} + \mathbf{c}')$	analytic (Eq. 13)	LO of Eq. (13)
Thermodynamics	from $R(\mathbf{c})$ & GCE	from $R(\mathbf{c})$ & GCE	Eqs. 19–32	GCE at μ^*
Exact conservation	up to quad. error	broken at $\mathcal{O}(V_c^{-1})$	exact by construction	exact by construction
Finite- μ stability	contour shift	native	native	native
Large- V_c stability	degrades	stable	stable	stable
Small- V_c validity	exact	breaks down	breaks down	breaks down
<i>Timing (full HRG, ~ 500 species, $V_c = 5000 \text{ fm}^3$):</i>				
Means only	~ 10 ms	~ 20 ms	~ 20 ms	~ 20 ms
With fluctuations	$\sim 18\text{--}140$ s	~ 1.5 s	~ 1 s	< 1 s

When to use which.

- **GaussLegendre**: Small to moderate volumes ($V_c \lesssim 100\text{--}1000 \text{ fm}^3$) or when the saddle-point approximation is not reliable. The saddle-point contour shift enables use at finite μ for partial canonical ensembles. Increase the multiplier m for better precision.
- **SaddlePoint**: Large volumes where GL suffers from cancellation. Best when exact partition function ratios $R(\mathbf{c})$ are needed (e.g. for higher-order cumulants beyond covariances).
- **SaddlePointNLO**: Large volumes when exact conservation of mean charges matters (e.g. thermal fits). Fastest for fluctuations. Does not compute individual $R(\mathbf{c})$ values.
- **SaddlePointLO**: Large volumes when only leading-order results are needed. Mean yields and thermodynamics coincide with GCE at μ^* , but fluctuations include canonical suppression ($-W_l^{(2)}W_m^{(2)}G_{lm}$ term). Useful as a baseline or for non-ideal models where NLO corrections have not been derived.

References

- [1] V. Vovchenko, “Saddle-point approximation for the canonical partition function: cumulant generating function, means, correlations, and fluctuations”, companion note (2025).
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